

Motion Prediction based Safety Boundary Study in Virtual Reality

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ABSTRACT

Nowadays, VR technology has played an indispensable role in promoting the construction of the Metaverse. In this study, we tried to avoid users' body physical collisions within the safety boundary in VR scenes. Three basic motion event categories are defined according to the common motions of users. We segmented users' data into three-dimensional coordinates, extracted the motion range features, and constructed the relational data set. Furthermore, we used statistical methods to study the correlation between user characteristics and motion range categories, in order to explore and analyze the possibility of optimizing the safety motion range. Finally, we deployed our safety boundary systems in the virtual reality environment.

Index Terms: User experience and usability—User Experience—;—Human factors and ergonomics—Safety Boundary

1 INTRODUCTION

With the improvement of VR hardware devices (such as higher resolution for VR headsets) and software upgrades (such as more accurate tracking and positioning systems with lower latency), users can experience more realistic virtual scenes. However, while having a deeper sense of immersion in the virtual world, users are likely to lose the perception of the surrounding physical environment, which may lead to colliding with objects around them, causing damage to the equipment and even users' body [1] [6]. The purpose of this research is to explore the relationship between the basic characteristics of users and their motion range in a specific VR scene. It can help VR software developers dynamically adjust the safety boundary, so as to avoid collision with real-world objects when users operate the VR devices.

2 BACKGROUND

Currently, the main input devices for human motion feature analysis are monocular cameras and multi-ocular cameras. The output data format is usually a collection of pixels that make up the image and video, and the human objects that need to be processed are generally presented in two-dimensional form [7]. In a VR scenario, an area needs to be delimited as a safety zone before users operate VR devices, so as to ensure that users are safe when using VR to prevent body collisions with objects in the real environment. Now, some researchers have made some progress in solving the problem of collisions within safety zones [2] [3] [4]. Their method effectively improved the space utilization rate in multi-user virtual reality scenarios [5].

3 VR MOTION EVENTS DEFINITION

Based on a large number of analyses on the position of the rhythm block and the users' body movements, we summarize and define

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three main types of VR motion events based on user motion features (in Fig.1):

- a) Slashing in front with one hand
- b) Slashing beside with one hand
- c) Slashing backhand

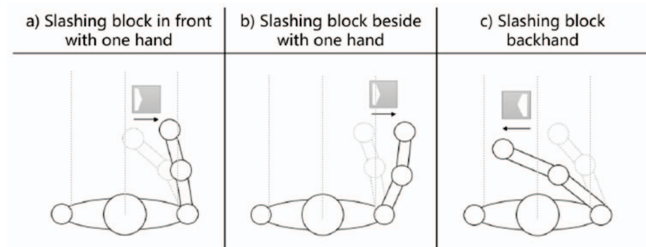


Figure 1: Definition of VR Motion Events

These types of motion events are typical, representative, and have obvious differences in the range of amplitude. As shown in the Fig.1, we set the plane formed by the user's shoulder and head as the reference system plane. Draw a 90° axis according to the right shoulder and define it as the dividing line of events (a) and (b). When the block flies to the user, if it is located on the side near the head at the dividing line, it will be counted as an event (a); If the block is located on the side away from the head, it is counted as event (b). Event (c) is a reverse cutting action. The plane of the reference system is consistent with the above. At this time, the dividing line takes the head as the 90° axis. Blocks to the left of the dividing line that require the user to reverse cut will be classified as event (c). It is worth mentioning that we only focus on the motion amplitude range of the user's motion under the horizontal plane to solve the collision problem. Therefore, if a user makes a right cutting action and is classified as the event (a), it means that the user may have made one of the right horizontals, right up, and right down cuts.

4 EXPERIMENT AND RESULTS

In the experiments, participants of 62 people (34 males, 28 females; height: $M = 166.9\text{cm}$, $SD = 8.96$) aged 18-23 ($M = 21.3$, $SD = 1.82$) took part in the study. They were instructed to begin the experiment for collecting motion data. Throughout the experiment, we conducted VR experiments in different sites (shown in Fig.2, Fig.3 and Fig.4) with a unified scene in the VR environment—"Open Saber".

After obtaining the location of the motion range boundary, we sorted and processed the data and got the specific motion range, range category, and parameterized identification corresponding to each motion event. The results are shown in the Table 1.

Moreover, we perform Spearman Correlation Analysis based on previous data. As shown in Table.2, we can observe that the user's age and proficiency in games have a positive correlation with the VR experience and the user's range of motion, and the impact is more

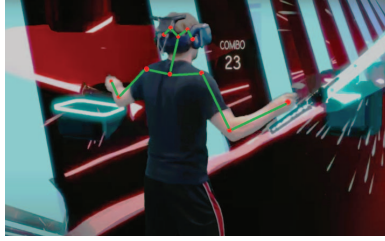


Figure 2: Slight slashing



Figure 3: Great slashing

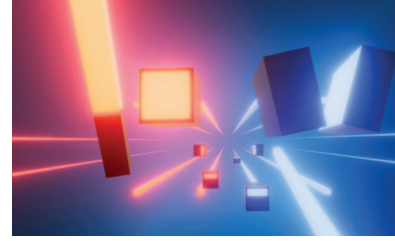


Figure 4: Experiment Environment

Table 1: Table of Motion Range and Range Category

Motion Event	Motion Range	Category	Category ID
a	[0, 0.397)	Small	0
	[0.397, 0.450)	Medium	1
	[0.450, 0.507)	Large	2
	[0.507, +∞)	Very Large	3
b	[0, 0.403)	Small	0
	[0.403, 0.453)	Medium	1
	[0.453, 0.503)	Large	2
	[0.503, +∞)	Very Large	3
c	[0, 0.412)	Small	0
	[0.412, 0.475)	Medium	1
	[0.475, 0.536)	Large	2
	[0.536, +∞)	Very Large	3

Table 2: Table of Spearman Correlation Analysis

Characteristics Features	Coefficient	Sig. (2-tailed)	Level
Age	0.216	0.001	High
Familiarity	0.211	0.001	High
Weight	-0.197	0.003	High
VR Experience	0.194	0.003	High
Height	-0.138	0.036	Medium

significant. The user's weight has a negative correlation with the range of motion, and the impact is more significant. The height of the user is negatively correlated with the magnitude of the motion, but the significant effect is general.

Supported by above results, the run-time data will be transmitted to our system to predict the three VR motion events, and the corresponding action amplitude categories will be output respectively. According to the three motion amplitude categories, the system will dynamically adjust the safety boundary according to the preset motion event occurrence period. As shown in Figure 5, to distinguish between dynamic and static safety boundaries during actual project deployment, we use orange as the color for dynamic safety boundaries and blue (the default) as the color for static safety boundaries.

5 CONCLUSION

With VR technology developing, there are still many problems to be solved. The purpose of this study is to solve the collision problem at the edge of the safety boundary and explore and study the relationship between user characteristics and the range of user motion in a VR environment. The experiment can still be further improved. First, for the recognition of VR motion events, we have adopted manual segmentation of the same game, and manually divided the motions made in a specific period into a certain VR motion event, which



Figure 5: Static and Dynamic VR Safety Boundary Deployment

can be generalized. Second, our data sample needs to be expanded. Due to the impact of the epidemic, we recruited a small number of subjects and obtained a limited number of experimental samples. Third, the correlation information between the basic characteristics of the user and the range of motion still needs to be studied. The implicit correlation contained in user features has not been fully explored, and follow-up research can use other methods to perform information mining on features in different dimensions.

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